Field report for Tavily's engineers

Shadow CPI, built for the Best Use of Tavily prize. Numbers read from the committed results files by `python tools/sync_doc_numbers.py` (as of 29 Oct 2026); edit `docs/templates/field_report_for_tavily.md`, not this file.

Conflict of interest: this hackathon is hosted by Nebius, which owns Tavily; see the README.

The headline measurement

- With `start_date`/`end_date` and `filter_by_published_date`, 0 of 42 cases detected leaking (Wilson 0–8%); pre-registered pilot plus replication.
- With no dates, 23 of 42 (Wilson 40–69%); a lower bound, because a leak counts only when the Wayback Machine confirms it.
- The zero is not proof of no leak: the archive confirmed 64 of 114 time-locked results as published before the cutoff, 50 could not be checked, and a content check found 1 confirmed post-cutoff fact in 55 sampled excerpts (exploratory).

Three questions, Tavily answers two

1. **Which pages?** Dates choose pages: `start_date`/`end_date` select by `published_date`, and 532 of 532 dated results fell inside the window we sent. Answered.
2. **How old is the cached copy?** `max_age_hours` is cache age, a freshness bound. Answered.
3. **Which version of the page existed on the cutoff date?** Nobody answers this. `R196` (TCS results preview) was dated 7 Jul, before its 8 Jul cutoff, and updated on 9 Jul with a link to the actual result; Extract returns the updated page. The content check found the same thing inside a page that passed the filter: a related-stories item dated after the cutoff (`results/content_check_labels.json`). `extract_as_of` fills the gap from archives in our code (demo, n=1; `results/asof_demo_r196.json`).

Field notes

Each note has observed / expected / suggested change, request IDs and a one-request repro script: `docs/tavily_bug_reports.md`. Re-check of 27 Oct 2026 (`results/repro_check_2026-10-27.json`): 5 of 7 reproduced, 1 did not, 1 unclear. Verdicts say what showed on that date only; the original measurements stand as measured.

#	Note	Tag against the docs	Re-checked 27 Oct 2026	Repro script
1	An as-of / crawl-timestamp field: date filters select which pages, not which version	documented behaviour	reproduced	`repro/extract_no_asof.py`
2	Separate original-publication and last-modified dates	documented behaviour	unclear	`repro/published_vs_modified.py`
3	auto_parameters picked a recent time_range for queries about a past month	unexplained	not reproduced	`repro/auto_parameters_time_range.py`
4	With explicit dates, the auto_parameters echo still reports a time_range	doc gap	reproduced	`repro/auto_parameters_echo_dates.py`
5	Map returns empty results silently	unexplained	reproduced	`repro/map_empty.py`
6	One scanned PDF: Extract returned the letterhead but not the image-only rates	doc gap	reproduced	`repro/scanned_pdf_extract.py`
7	/usage lags per-call usage	doc gap	reproduced	`repro/usage_lag.py`

Three feature requests

1. **Say where `published_date` comes from and how confident it is.** The documented definition is published or last updated (beta). It was later than the page's own date on 4 of 42 distinct date-test pages, and a caller cannot tell which kind of date a result carries. Suggested: a source field (page metadata, sitemap, initial crawl) and a confidence value; with a source field, the two separate dates in note 2 follow. Evidence: `results/date_field_test.json`.
2. **An as-of / snapshot mode for Extract and Search excerpts.** Return the copy as it was on a given date, or at least the crawl time of the copy returned. Dates select pages, not versions: `R196` was dated 7 Jul, before

its 8 Jul cutoff, and updated on 9 Jul with a link to the actual result; Extract returns the updated page, and `max_age_hours` bounds cache age, not a date. Evidence: `results/timelock34_second_review.json`, `results/asof_demo_r196.json`.

3. **Date parameters on `/research`.** The request body has no date parameter (checked 29 Oct in the API reference), so a research call cannot be time-locked the way Search can; we do not use it for dated work for that reason. Not measured. Evidence: `config/doc_citations.json`.

Reproduce

`repro/README.md`: one request per note, your own key, at most a few credits each. Every API call is logged in a hash-chained log (`results/audit_log/`).